

Japanese beetle



Scientific classification

Kingdom: [Animalia](#)
Phylum: [Arthropoda](#)
Class: [Insecta](#)
Order: [Coleoptera](#)
Family: [Scarabaeidae](#)
Genus: [Popillia](#)
Species: ***P. japonica***

Binomial name

Popillia japonica

[Newman](#), 1841

JAPANESE BEETLE

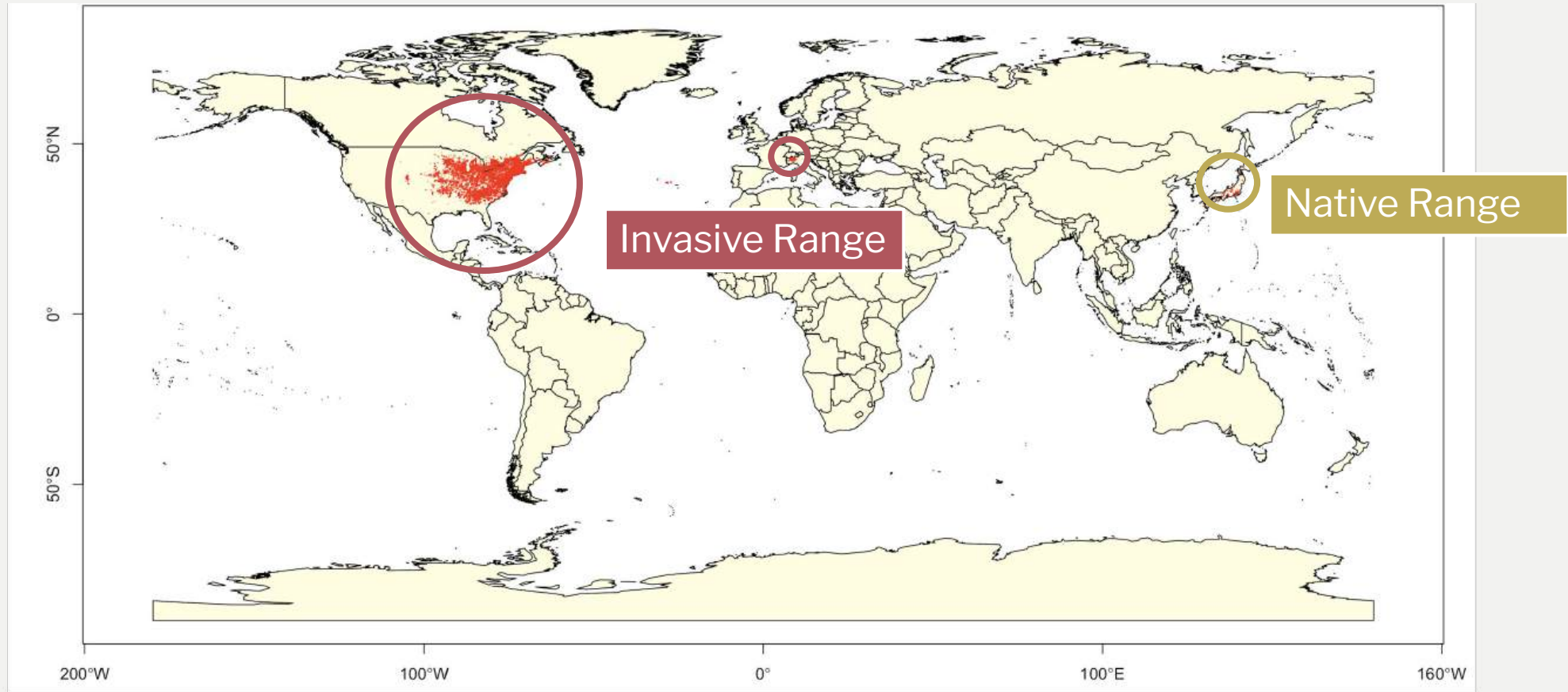
(Popillia japonica)

Habitat Distribution Modeling for an Invasive Species

Ecology and Biology

- It over-winters as larvae
- Adults emerge in greatest numbers during early July
- Most active: warm, sunny days
 - ***Temperature of 21°C***
 - ***Relative humidity of 60%***
- Activity (flying) threshold
 - ***T°<35°C***
 - ***Humidity<60%***
- Moisture is crucial to egg hatch and larval development.

Global Distribution



Observation data source: iNaturalist

Habitat Suitability Modeling

Using Maxent (Maximum Entropy) modeling

Model 1:

Training data set: native observation points

Testing data set: invasive observation points

Model 2:

Training data set: 80% of all observation points

Testing data set: 20% of all observation points

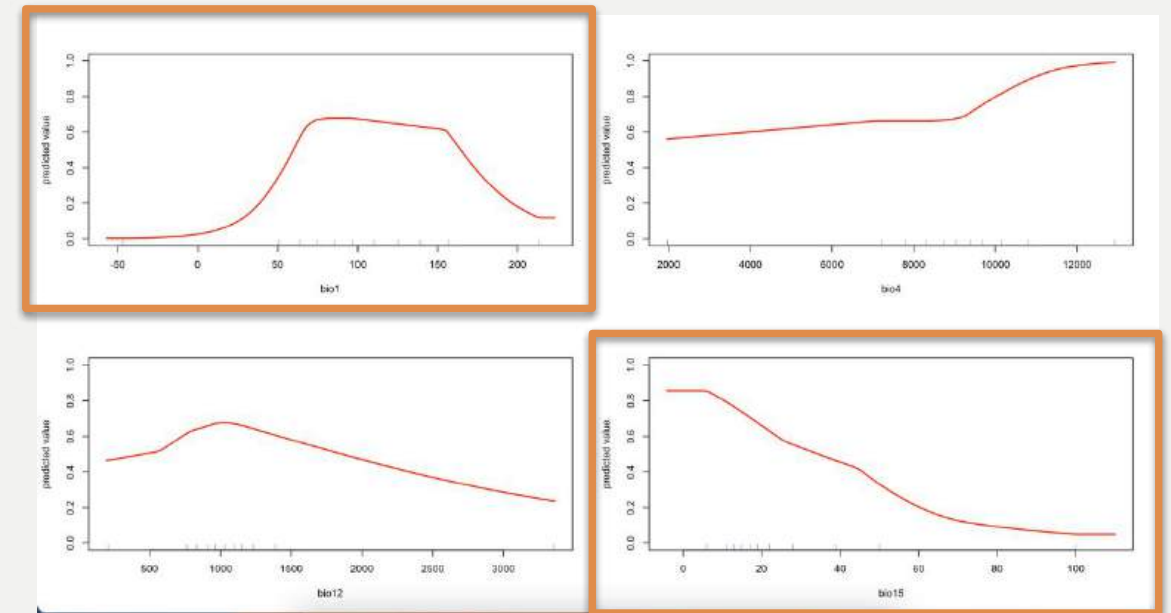
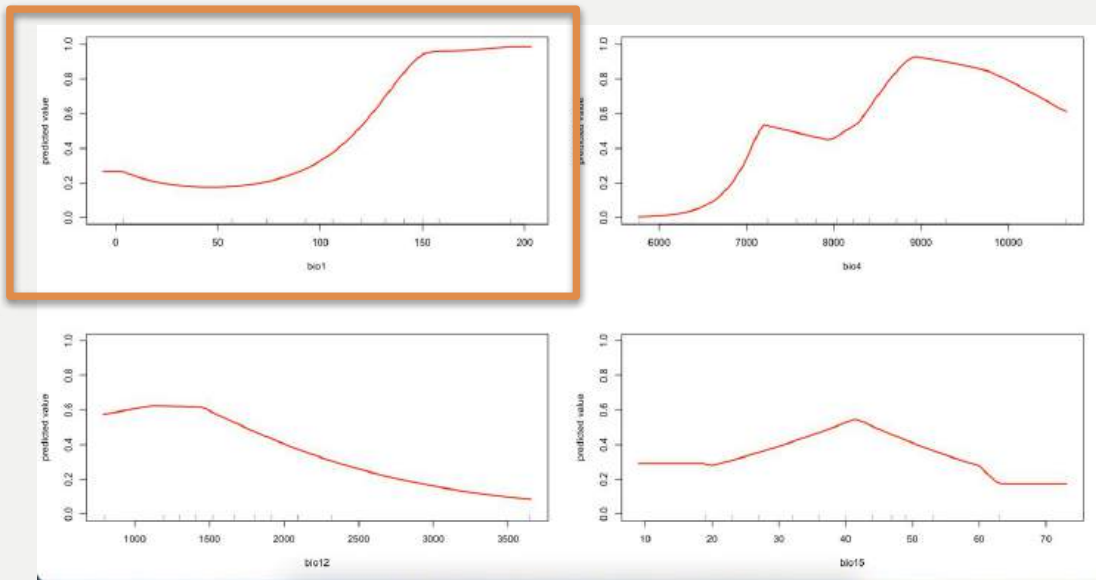
Bioclimatic Variables Responses

Model 1:

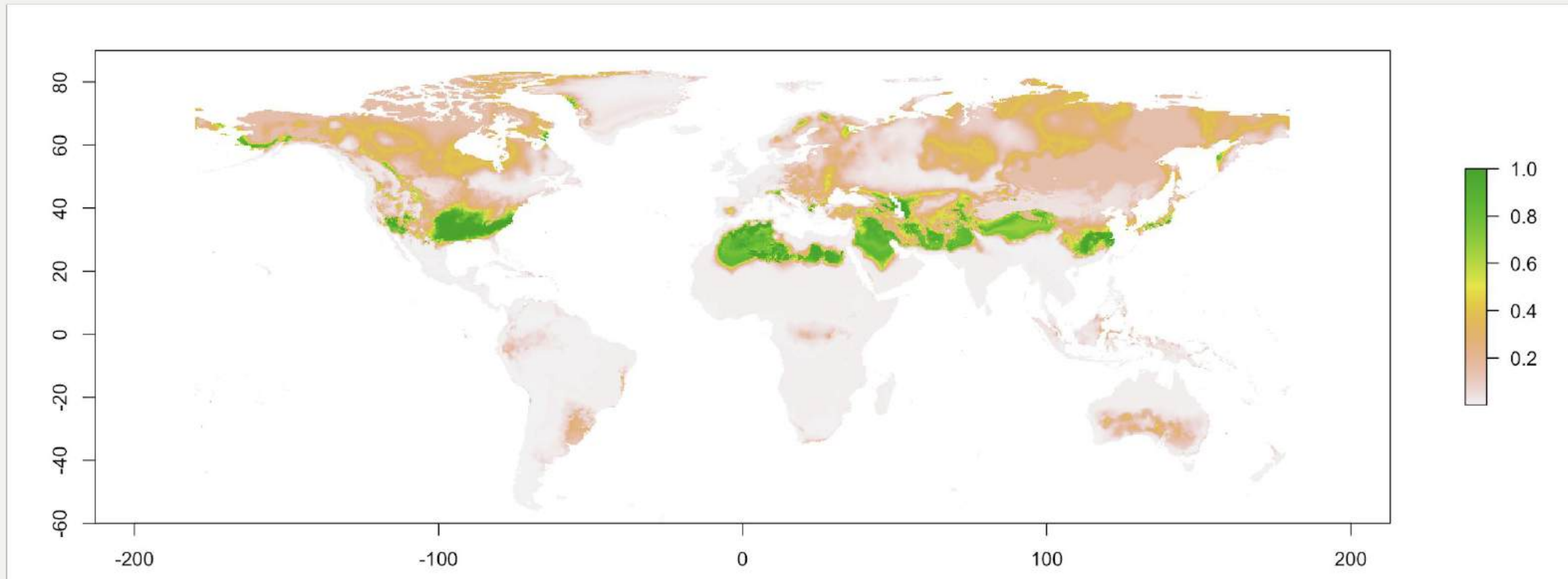
Variable	Percent contribution	Permutation importance
bio1	46.2	52.9
bio15	24	17.8
bio12	16.5	10
bio4	13.3	19.3

Model 2

Variable	Percent contribution	Permutation importance
bio1	39.7	39.5
bio15	39.7	43.6
bio12	13.7	4.3
bio4	6.9	12.5

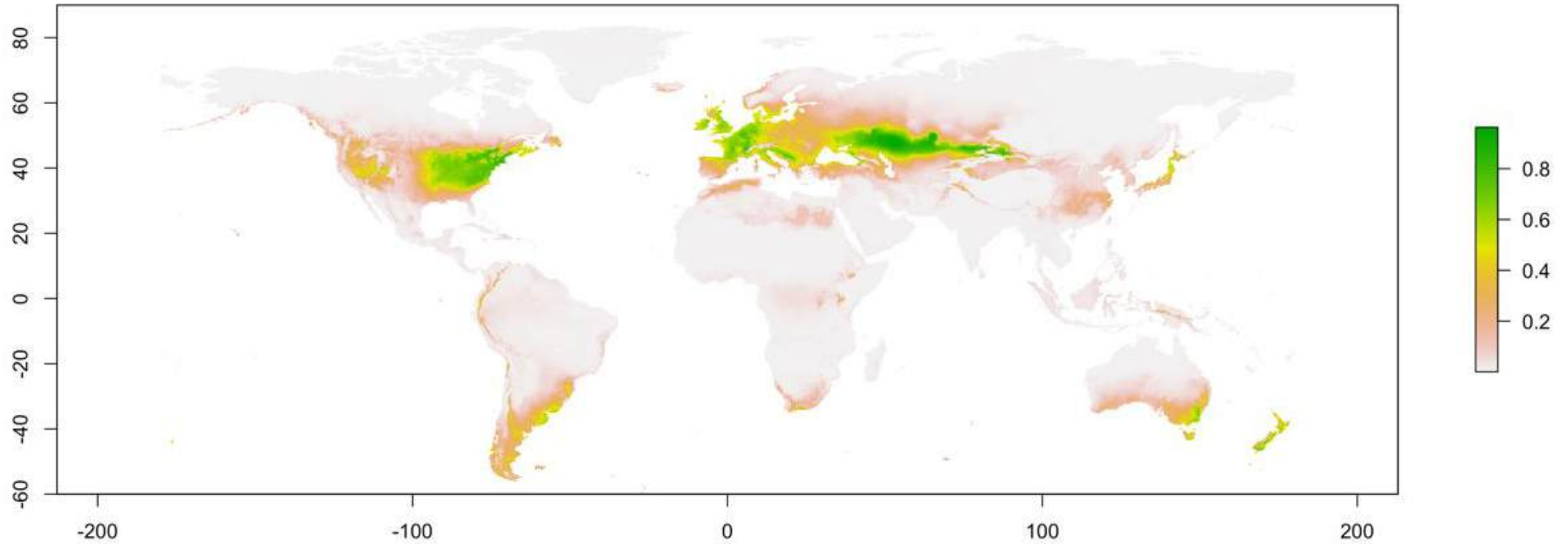


Suitability Map by **Model 1**



AUC = 0.468
Not a reliable model.

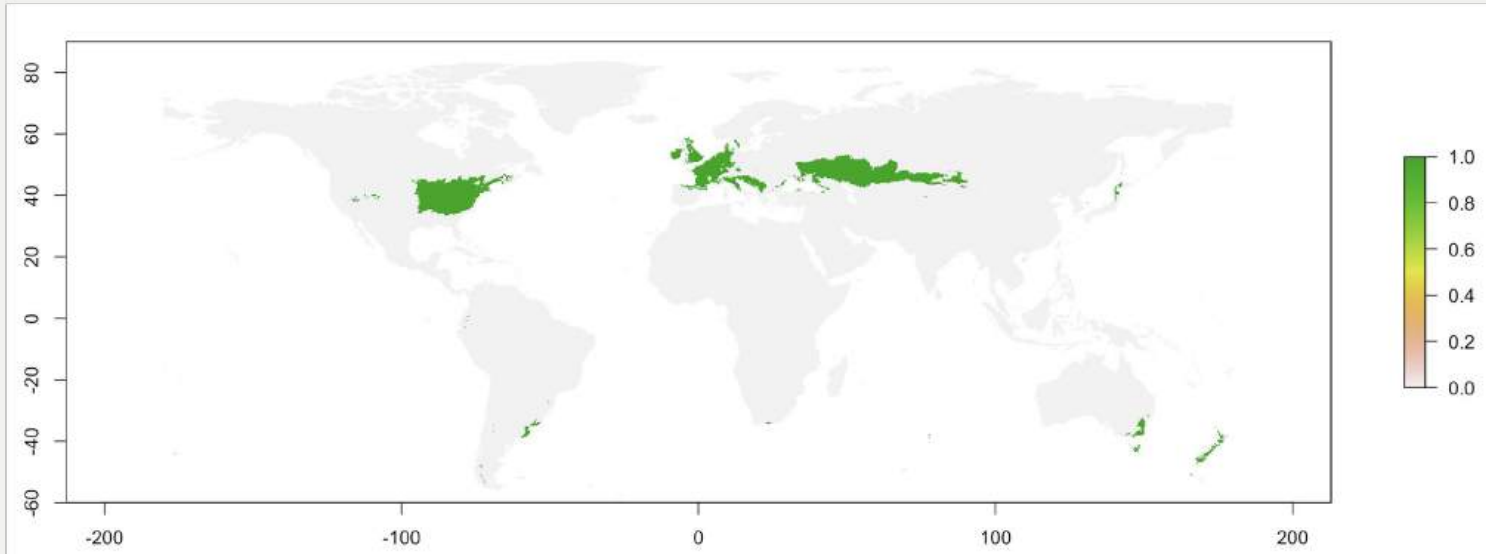
Suitability Map by **Model 2**



AUC = 0.71

It is a OK model, though not reliable enough.

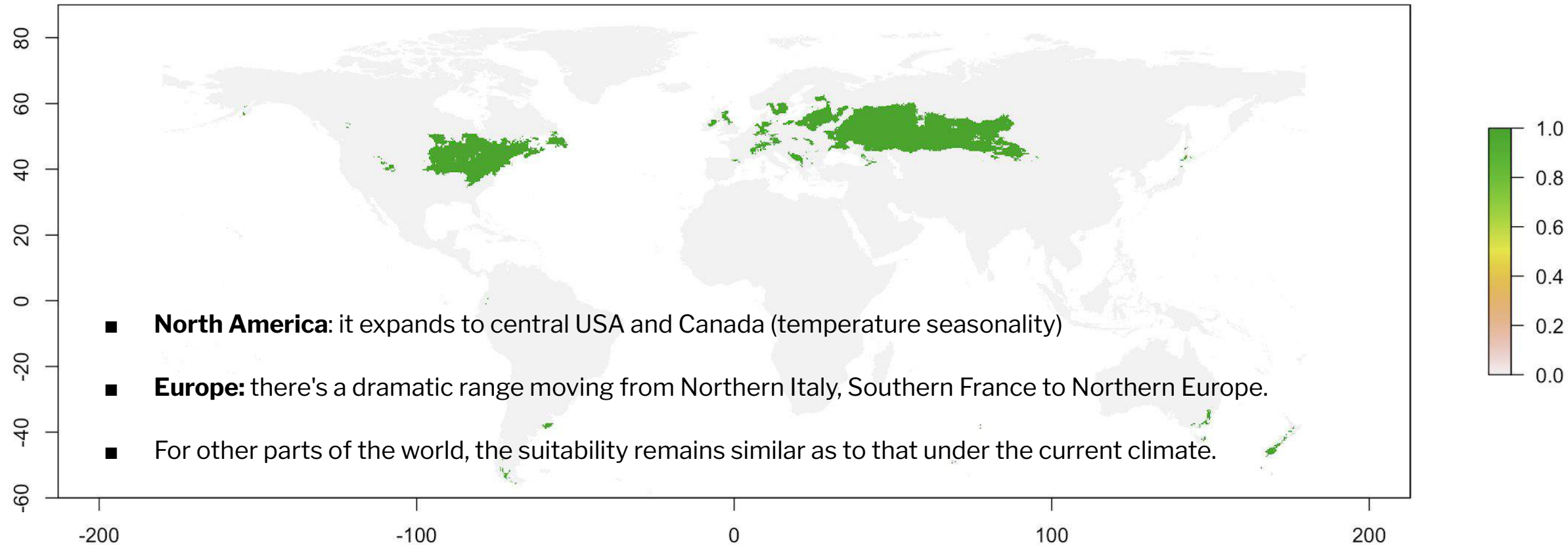
Distribution Prediction under current climate



- Eastern USA
- Northern Italy (more precisely Milano - Lombardy area)
- Central Europe can potentially be suitable for this species to invade (similar climates as Milano's)
- It shows that this species may start to invade the low latitudes.

Distribution Prediction under future climate*

*RCP 4.5 greenhouse gas concentration trajectory

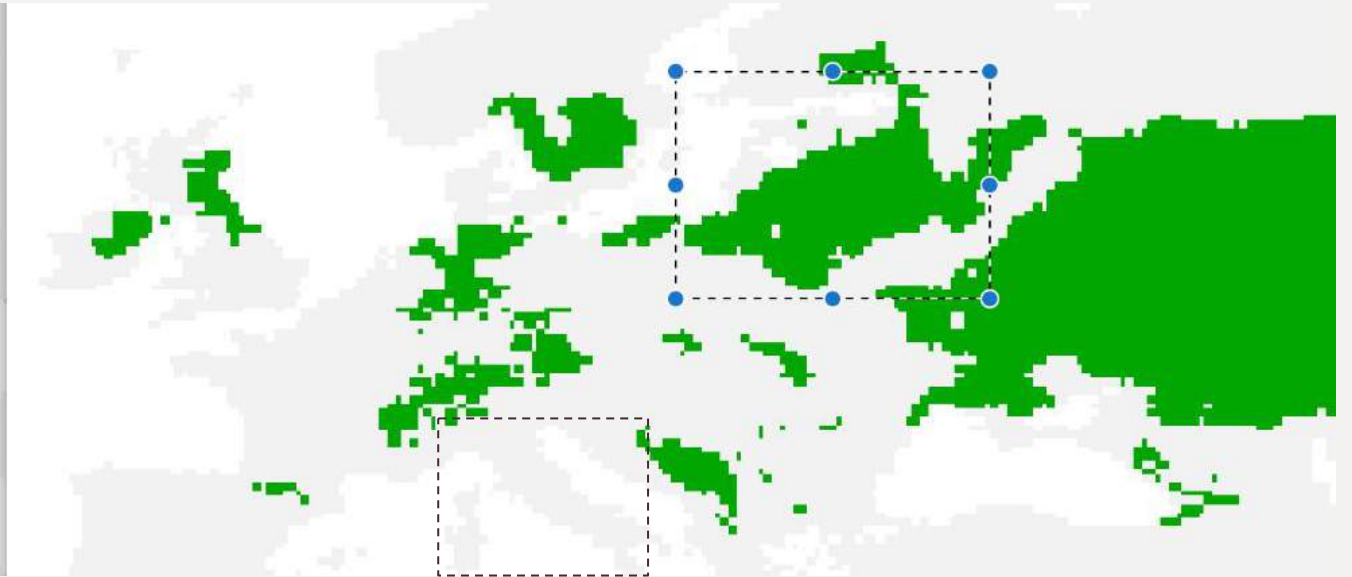


Distribution Prediction in Italy

based on current climate and future climate



Current Climate



Future Climate (RCP 4.5)

The Japanese Beetles will move out of Italy.

Thank you!

Jensen Sentao Zeng, Tamara Chwojnik, Diabou Mbaye
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